EGYPT-JAPAN UNIVERSITY OF SCIENCE AND TECHNOLOGY (E-JUST) — SCHOOL OF ECCE

# A Literature Review On Multi-Modal Edge Sensing For Post-Harvest Fruit Spoilage Detection

*ECE 315: Seminar and Project-Based Learning in ECE • Fall Semester 2026 • Supervisor: Dr. Adel Bedair*

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***Abstract*—Post-harvest decay of fresh fruit causes major economic loss and food waste, yet most automated freshness monitors rely on a single sensing modality. This review organizes the literature on non-destructive spoilage detection into five themes: machine vision, electronic-nose (e-nose) gas sensing, low-cost IoT monitors, multi-sensor data fusion, and edge-deployable learning. Comparing these themes shows that vision detects only surface symptoms, gas sensing is blind to produce species and suffers from drift, and existing fusion studies are mostly benchtop and offline. The resulting gap is a low-cost, continuously operating system in which vision supplies produce context to a drift-normalized gas channel, enabling species-aware early warning of hidden decay.**

***Index Terms*—fruit spoilage detection, electronic nose, computer vision, multi-modal sensor fusion, edge AI, IoT.**

## I. Introduction

Post-harvest decay during storage, transport, and retail display causes substantial financial loss and food waste. Quality is still judged mainly by manual inspection, which is slow, subjective, and unable to see decay that starts inside the fruit before the peel changes [1]. This section groups prior work into five themes (Sections II-A to II-E), compares what the studies agree on and leave unsolved, and derives a single research gap (Section III) that motivates the multi-modal edge system proposed in this work.

## II. Literature Review

### A. Machine-Vision Inspection

Cubero *et al.* [1] showed that RGB machine vision reliably grades external attributes such as color, size, shape, and surface defects, provided illumination is tightly controlled, and noted that internal quality requires non-visible techniques. Deep learning has since become dominant: surveying 40 agricultural studies, Kamilaris and Prenafeta-Boldú [2] found that CNNs outperform conventional methods but depend on large labeled datasets and generalize poorly to unseen conditions. Kazi and Panda [3] reported over 99% fresh-versus-rotten accuracy with residual CNNs, but on curated images with clean backgrounds. These studies agree that vision can tell *which* fruit is present and whether its surface is damaged, but not whether decay has begun inside.

### B. Electronic-Nose Gas Sensing

Baietto and Wilson [4] showed that volatile organic compound (VOC) profiles change measurably with fruit ripening and decay, and that low-cost metal-oxide semiconductor (MOS) sensors are the most common sensing element. Because these gases are released before visible damage, gas sensing can warn earlier than a camera. Two weaknesses limit this. First, climacteric fruits (banana, apple) emit an ethylene burst during normal ripening while non-climacteric fruits (orange, grape) emit very little [5], so the same reading can mean ripening for one species and spoilage for another. Second, Vergara *et al.* [6] recorded significant drift in a 16-sensor MOS array over 36 months; it could be compensated only with periodic recalibration data, which an unattended low-cost node rarely has. MOS readings are also sensitive to temperature and humidity [4].

### C. Low-Cost IoT Spoilage Monitors

Bhattacharjee *et al.* [7] place MQ-3, MQ-4, and MQ-135 gas sensors and a DHT22 probe on a sealed container read by an ESP32, log the data to the cloud, and map readings to five grades with a weighted quality index. The system costs about USD 18 and its sealed container concentrates volatiles. However, its thresholds were set from visual cues for one fruit (banana) in one trial, with no learning, no drift compensation, and no accuracy reported against ground truth [7]. Table I compares this baseline with the other themes.

TABLE I
COMPARISON OF LITERATURE THEMES

| Theme | Main strength | Main limitation |
|---|---|---|
| Machine vision [1]–[3] | Accurate on surface defects; identifies produce | Misses internal decay; lighting- and dataset-dependent |
| E-nose / MOS gas [4]–[6] | Detects VOCs before visible damage | Species-blind; drift and humidity sensitivity |
| IoT monitor [7] | Low cost; sealed container; dashboard | Fixed thresholds; one fruit; no accuracy data |
| Data fusion [8] | Fused instruments beat single ones | Mostly benchtop and offline |
| Edge learning [9] | Small models run on MCUs | Not validated on multi-modal spoilage |

### D. Multi-Sensor Data Fusion

Because vision and gas sensing fail in complementary ways, studies have combined them. Calvini and Pigani [8] reviewed fusion of electronic noses, tongues, and eyes at low (raw data), mid (features), and high (decision) levels, and found that fused systems generally outperform single instruments. Most reviewed studies, however, used laboratory instruments with offline analysis, and treated the modalities as parallel inputs; none used vision to supply *context*, such as produce species, that changes how the gas signal is interpreted.

### E. Edge-Deployable Learning

TensorFlow Lite Micro [9] runs quantized neural networks on microcontrollers with only kilobytes of memory, making on-device inference feasible. It was validated on generic benchmarks rather than spoilage data with concurrent sensors, and on an ESP32-CAM the frame buffer and Wi-Fi stack already compete for memory. A node that acquires data in a sealed chamber (Fig. 1) and sends it to a nearby inference server is therefore a practical intermediate step, and is the approach adopted in this work.

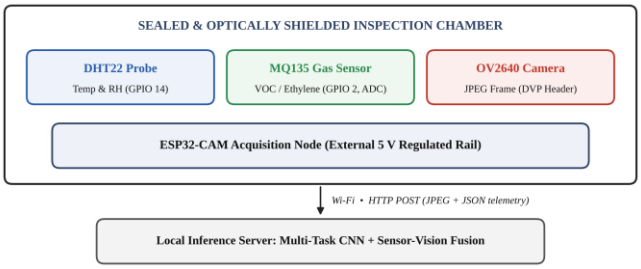



Fig. 1. Multi-modal sensing chamber: the ESP32-CAM acquires gas, climate, and image data and sends them to a local inference server.

## III. Research Gap and Positioning

The themes reveal complementary gaps: vision is limited to the surface and its training conditions [1]–[3]; gas sensing is early but species-blind and drift-prone [4]–[7]; fusion is benchtop, offline, and symmetric [8]; and edge models have not been applied to multi-modal spoilage monitoring [9]. No reviewed work provides a low-cost, continuously operating system in which vision identifies the produce species and selects species-specific, drift-normalized gas thresholds, so that hidden decay is flagged before visible symptoms appear.

The proposed system targets this gap with an ESP32-CAM, MQ135, and DHT22 inside a sealed, optically shielded enclosure (Fig. 1). A multi-task CNN reaches 92.3% produce and 95.1% freshness validation accuracy (89.4% joint exact-match on the test set). The gas reading is normalized to a per-session baseline ratio $G = R/R_0$ to reduce drift [6], the detected produce selects a produce-specific gas limit, and decision-level fusion outputs FRESH, EARLY WARNING, or SPOILED. On-device quantized inference [9] and multi-day validation across species remain future work.

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